

Energy Ledgers for Forced Harmonic ODEs

Kirchhoff power balance and an unknown component with $X = LRC$

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Abstract

Multiplying a forced harmonic LRC equation by current turns its voltage balance into a power balance. Opposite entries identify the heat and source equations needed for a complete energy ledger; their remaining terms are left open. We then add the fictional voltage term Xq''' , with $X > 0$ and its physical identity unknown. The sign of its power determines whether it supplies or absorbs energy. Exact numerical examples show both cases. Assuming only the coefficient relation $X = LRC$, we check the units and compare how component choices strengthen either contribution to the circuit's power equation. A fourth part carries the same model to a rotary impact driver and compares its fictional contribution with ordinary loss reduction, mechanical design and control.

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1 Kirchhoff's voltage law and the missing accounts

Consider the forced harmonic equation for a series LRC circuit,

$$Lq'' + Rq' + \frac{q}{C} = f(t), \quad L > 0, \quad C > 0, \quad R \geq 0. \quad (1)$$

Here q is charge, L is inductance, R is resistance, C is capacitance, and $f(t)$ is the applied voltage. Coefficients are constant and trajectories are smooth. The equation is second order because its highest derivative is q'' ; it is linear and has degree one in that derivative.

Every term in (1) is a voltage. The component laws are $v_L = Lq''$, $v_R = Rq'$, and $v_C = q/C$. Kirchhoff's voltage law, written with the source first, is

$$f(t) - Lq'' - Rq' - \frac{q}{C} = 0. \quad (2)$$

Once the ODE is solved, we have $q(t)$ and $q'(t)$. The series current is $i = q'$. Multiplication by that current turns the voltage law into a *power law*:

$$f(t)i - Lq''i - Rq'i - \frac{qi}{C} = 0. \quad (3)$$

Each term is now voltage times current, measured in watts. The forcing contributes $+fi$ to this equation; resistor absorption contributes $-Rq'i = -Ri^2$.

The power sum is zero. To relate it to constant total energy, include the heat recipient and the forcing source or sink:

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0. \end{cases} \quad (4)$$

The first line belongs to the heat recipient and the third to the forcing source or sink. Their full ODEs are unknown, but one transfer term in each is already determined. The heat entry $-Rq'i$ in line 2 has the matching $+Rq'i$ in line 1. The source entry $+fi$ in line 2 has the matching $-fi$ in line 3.

This is the energy counterpart of double-entry economic accounting: a transfer debits one account and credits another. If $fi < 0$, the forcing source receives energy, and those entries reverse direction.

The cancellation identifies the transfers needed for a complete ledger. A proof that total energy is constant also requires the omitted terms, including the relevant energy rates and physical reservoir laws. We leave those blanks open here; completing them is a separate article. A prescribed $f(t)$ alone does not specify a source's available energy.

2 Does the unknown term Xq''' supply or absorb power?

2.1 The fourth line and the direction of transfer

Add one fictional component term, with constant $X > 0$:

$$Xq''' + Lq'' + Rq' + \frac{q}{C} = f(t). \quad (5)$$

Its physical identity is unknown. Its proposed voltage contribution is $v_X = Xq'''$. The power equation and its matching entries become

$$\begin{cases} 1 : & \dots + Rq'i + \dots = 0, \\ 2 : & f(t)i - Xq'''i - Lq''i - Rq'i - qi/C = 0, \\ 3 : & -f(t)i + \dots = 0, \\ 4 : & \dots + Xq'''i + \dots = 0. \end{cases} \quad (6)$$

The fourth line carries the opposite of the new contribution in line 2. Define the two signed powers explicitly:

$$P_X = Xq'''i \quad (\text{received by } X), \quad P_{X \rightarrow 2} = -Xq'''i \quad (\text{contribution to line 2}). \quad (7)$$

Table 1. Direction follows from the derivative product because $X > 0$.

Sign of $q'''i$	P_X	$P_{X \rightarrow 2}$	Role of X
Negative	Negative	Positive	Supplier
Positive	Positive	Negative	Absorber
Zero	Zero	Zero	No instantaneous transfer

For example, positive current together with negative q''' makes X a supplier. Positive current together with positive q''' makes it an absorber. A positive coefficient alone does not select either role.

Energy transfer follows from the signed power integral:

$$W_X(t_0, t_1) = \int_{t_0}^{t_1} X q'''(t) i(t) dt, \quad W_{X \rightarrow 2} = -W_X. \quad (8)$$

Positive W_X is net energy received by X ; negative W_X is net energy supplied by it. This specifies the required transfer while leaving the remaining terms of line 4 open.

2.2 Two numerical examples from exact trajectories

Take the illustrative values $L = 1 \text{ H}$, $R = 1 \Omega$, $C = 1 \text{ F}$, and $X = 1 \text{ H s}$. Put $u = t/(1 \text{ s})$. In each example choose $q(t)$ and obtain the required $f(t)$ by substitution into (5).

Supplier. Choose

$$q(t) = (1 \text{ C}) \sin u, \quad i(t) = (1 \text{ A}) \cos u, \quad q'''(t) = -(1 \text{ A/s}^2) \cos u. \quad (9)$$

The inductive and capacitive voltages cancel, as do the resistive and X voltages, so the required forcing is $f(t) = 0$. The two powers are

$$P_X = -(1 \text{ W}) \cos^2 u, \quad P_{X \rightarrow 2} = +(1 \text{ W}) \cos^2 u. \quad (10)$$

At $t = 0$, X contributes $+1 \text{ W}$ to line 2. Over the exact period $T = 2\pi \text{ s}$,

$$W_X = -(1 \text{ J}) \int_0^{2\pi} \cos^2 u du = -\pi \text{ J}. \quad (11)$$

Thus X supplies $\pi \text{ J}$ during the cycle. Here $P_X = -Ri^2$ pointwise, so the required transfer pays for the resistor's absorption. The equation identifies the required debit in the unknown account; it does not specify what physically supplies that energy.

Absorber. On $0 \leq t \leq 1 \text{ s}$, choose

$$q(t) = (1 \text{ C}) \frac{u^3}{6}, \quad i(t) = (1 \text{ A}) \frac{u^2}{2}, \quad q'''(t) = 1 \text{ A/s}^2. \quad (12)$$

Direct substitution gives

$$f(t) = (1 \text{ V}) \left(1 + u + \frac{u^2}{2} + \frac{u^3}{6} \right). \quad (13)$$

Now $P_X = (1 \text{ W}) u^2/2 > 0$ for $t > 0$, so X absorbs power. At $t = 1 \text{ s}$, its contribution to line 2 is $P_{X \rightarrow 2} = -1/2 \text{ W}$. Its received energy is

$$W_X = (1 \text{ J}) \int_0^1 \frac{u^2}{2} du = \frac{1}{6} \text{ J}. \quad (14)$$

The same positive X therefore supplies energy on the sinusoid and absorbs energy on the polynomial trajectory. Both examples are exact substitutions and evaluated power integrals.

3 Choosing L , R and C when $X = LRC$

3.1 Units and what the coefficient determines

Assume only that the numerical value of the coefficient is given by

$$X = LRC > 0. \quad (15)$$

Consequently $R > 0$ in this part. The identity and mechanism of X remain unknown. The relation determines its coefficient value, including units.

Since $\Omega F = s$, the dimensional check is

$$[X] = [LRC] = H s = \frac{V s^2}{A}, \quad [q'''] = \frac{A}{s^2}, \quad [Xq'''] = V. \quad (16)$$

Thus the new term is a voltage, and its contribution to line 2 is

$$P_{X \rightarrow 2} = -LRC q''' i. \quad (17)$$

3.2 Maximizing the supplying or absorbing contribution

First hold the trajectory $q(t)$ fixed, including the time scale and amplitude. Then the derivative product $q''' i$ is fixed at each instant. Equation (17) gives the two design cases directly:

Table 2. At fixed trajectory, both magnitudes increase with the product LRC .

Objective at an instant	Required trajectory sign	Magnitude to maximize
Supply power to line 2	$q''' i < 0$	$LRC(-q''' i)$
Absorb power from line 2	$q''' i > 0$	$LRC(q''' i)$

Increasing any of L , R , or C strengthens the existing direction of transfer. Positive component choices cannot reverse that direction for the same trajectory. With independent positive bounds on the three components, the largest product uses all three upper bounds. Without such bounds there is no finite maximum. The signed work in (8) scales in the same way when the trajectory and interval are fixed.

For the sinusoidal family $q = Q \sin(\omega t)$, where $Q > 0$ is charge amplitude and $\omega > 0$ is angular frequency,

$$q''' = -\omega^2 i, \quad P_{X \rightarrow 2} = LRC \omega^2 i^2 \geq 0. \quad (18)$$

Every positive choice of L, R, C therefore makes X a supplier on this trajectory, except at current zeros. Absorption requires a trajectory with $q''' i > 0$, such as the polynomial in (12); changing positive coefficients alone cannot produce it on the sinusoid.

If “effect” instead means magnitude relative to the resistor’s absorption, then, at $i \neq 0$,

$$\frac{|P_{X \rightarrow 2}|}{R i^2} = LC \left| \frac{q'''}{i} \right|. \quad (19)$$

At fixed trajectory, larger L or C increases this ratio. Larger R increases both powers equally and leaves the ratio unchanged. On a sinusoid the ratio is $LC \omega^2$.

3.3 Numerical comparison in both roles

Use the two trajectories from Part 2. Compare the supplier at $t = 0$ and the absorber at $t = 1$ s. For each component choice, recompute $f(t)$ from (5) so that the specified trajectory is retained.

Table 3. Exact signed contributions $P_{X \rightarrow 2}$ of the unknown term.

(L, R, C) in (H, Ω, F)	X (H s)	Supplier (W)	Absorber (W)
(1, 1, 1)	1	+1	-1/2
(1, 2, 1)	2	+2	-1
(2, 2, 2)	8	+8	-4

For example, with independent bounds $1 H \leq L \leq 2 H$, $1 \Omega \leq R \leq 2 \Omega$, and $1 F \leq C \leq 2 F$, the last choice maximizes both directional magnitudes for the specified trajectories. It supplies 8 W at the supplier instant and absorbs 4 W at the absorber instant. Over their respective intervals, the same exact integrals give 8π J supplied during the sinusoidal cycle and $4/3$ J absorbed during the polynomial interval.

These comparisons hold the motion fixed and adjust the forcing accordingly. If $f(t)$ and the initial conditions are held fixed instead, changing L, R, C also changes $q(t)$; maximizing the product alone then does not determine the maximum effect. The coefficient relation specifies the power required of X on a chosen motion. Its physical identity and the remaining terms of line 4 are still open.

4 The same equation in a rotary impact driver

4.1 A simplified output model

A rotary impact driver uses a motor to accelerate a hammer, which transfers torque to an anvil and then to the bit and fastener. Established hammer models distinguish acceleration, contact and release, with joint stiffness and losses affecting the blow (Wettstein et al., 2021). Here we retain one smooth loading interval within a blow.

Let $\theta(t)$ be the output's angular displacement from a fixed local reference, and $\Omega = \theta'$ its angular velocity. Use constant output inertia $J > 0$, resisting coefficient $b > 0$, and torsional stiffness $k > 0$. The resisting torques are $b\Omega$ and $k\theta$. The hammer applies a prescribed torque $\tau_h(t)$; its inertia is outside this output model and is not included in J . Torque balance gives

$$J\theta'' + b\theta' + k\theta = \tau_h(t). \quad (20)$$

This is the original harmonic ODE under the correspondence

$$q \leftrightarrow \theta, \quad i \leftrightarrow \Omega, \quad L \leftrightarrow J, \quad R \leftrightarrow b, \quad C \leftrightarrow 1/k, \quad f \leftrightarrow \tau_h.$$

The spring and damping laws define a local ideal model. Contact switching, successive blows and permanent fastener advance require additional laws.

Now assume that the fictional component exists somewhere inside the tool and acts at this output. Write its rotational coefficient as X_r :

$$X_r\theta''' + J\theta'' + b\theta' + k\theta = \tau_h(t), \quad X_r = \frac{Jb}{k} > 0. \quad (21)$$

The relation is the rotational counterpart of $X = LRC$. Since b/k has units of seconds,

$$[X_r] = \text{N m s}^3/\text{rad} = \text{kg m}^2 \text{ s}, \quad [X_r\theta'''] = \text{N m},$$

where radians are dimensionless in SI. Its coefficient is known; its physical mechanism remains unspecified.

Multiplication by Ω gives the same source-first power equation:

$$\tau_h\Omega - X_r\theta'''\Omega - J\theta''\Omega - b\Omega^2 - k\theta\Omega = 0. \quad (22)$$

The component receives $P_{X_r} = X_r\theta'''\Omega$, with the opposite entry in this equation. It supplies power when $\theta'''\Omega < 0$ and absorbs when $\theta'''\Omega > 0$. The heat and hammer accounts likewise receive $+b\Omega^2$ and $-\tau_h\Omega$, as in the earlier ledger.

4.2 Comparing the work on the same motion

Hold the trajectory and interval fixed and calculate the required hammer torque separately for the ordinary model and the model containing X_r . Call these torques τ_0 and τ_X . Their difference is $\tau_X - \tau_0 = X_r \theta'''$. Integrating their signed port powers gives

$$W_h^{(X)} - W_h^{(0)} = \int_{t_0}^{t_1} (\tau_X - \tau_0) \Omega dt = \int_{t_0}^{t_1} X_r \theta''' \Omega dt = W_{X_r}, \quad (23)$$

where each W_h is work delivered by the hammer to the output. Thus supplying reduces the required hammer work by exactly the work supplied by X_r . Absorbing increases it on the same motion; absorption can instead serve a braking objective when the derivative product has the required sign. Returning that absorbed energy later needs a component law.

For a concrete comparison, choose the smooth loading segment

$$\theta(t) = \Theta \sin(\nu t), \quad 0 \leq t \leq \frac{\pi}{2\nu}, \quad \Theta > 0, \quad \nu > 0. \quad (24)$$

The angle increases while the output slows to rest. All comparisons start with the same nonzero angular velocity $\Theta\nu$ and end at the same angle Θ . Since $\theta''' = -\nu^2 \Omega$, X_r supplies throughout the moving part. Its signed work and the heat transfer are

$$W_{X_r} = \int_0^{\pi/(2\nu)} X_r \theta''' \Omega dt = -\frac{\pi}{4} X_r \Theta^2 \nu^3, \quad (25)$$

$$W_b = \int_0^{\pi/(2\nu)} b \Omega^2 dt = \frac{\pi}{4} b \Theta^2 \nu. \quad (26)$$

On this trajectory, the required torque is also obtained by an ordinary model with reduced damping

$$b_{\text{eff}} = b - X_r \nu^2 = b \left(1 - \frac{J \nu^2}{k} \right), \quad b_{\text{eff}} \geq 0. \quad (27)$$

Indeed, $X_r \theta''' + b \Omega = b_{\text{eff}} \Omega$. This equality concerns the chosen motion. In the fictional model the original heat transfer remains and X_r supplies the difference. In the ordinary model less work becomes heat.

Take the following exact illustrative values; they are not measurements of a driver:

$$J = 10^{-4} \text{ kg m}^2, \quad b = \frac{1}{50} \text{ N m s/rad}, \quad k = 1000 \text{ N m/rad}, \quad \Theta = \frac{1}{10} \text{ rad}, \quad \nu = 1000 \text{ s}^{-1}.$$

Then $X_r = 2 \cdot 10^{-9} \text{ kg m}^2 \text{ s}$, $J \nu^2 / k = 1/10$, and $b_{\text{eff}} = 9/500 \text{ N m s/rad}$.

Table 4. Exact transfers; ΔW_h is the change in hammer work relative to the first model.

Model on the same motion	Heat (J)	W_{X_r} (J)	ΔW_h (J)
Ordinary, original b	$\pi/20$	0	0
Fictional, original b	$\pi/20$	$-\pi/200$	$-\pi/200$
Ordinary, b reduced by 10%	$9\pi/200$	0	$-\pi/200$

4.3 How much conventional effort can X_r replace?

An effective driver reaches a specified tightening target with less battery work and time, within torque, vibration and wear limits. Conventional mechanical design adjusts hammer preparation and contact with the anvil (Wettstein et al., 2021); control adjusts motor effort and impact timing (Benazet et al., 2026). The replacement by X_r can be stated more precisely at our output boundary.

During the useful motion, it can replace part of the hammer input. In the worked example, X_r supplies $\pi/200$ J, so the hammer can deliver exactly that much less work while retaining the same motion. Its assisting torque is $1/5$ N m at the start and falls to zero at the end. The ordinary hammer work, evaluated from its port power, is

$$W_h^{(0)} = \int_0^{\pi/(2\nu)} \tau_0 \Omega dt = \left(\frac{9}{2} + \frac{\pi}{20} \right) \text{J}. \quad (28)$$

Thus X_r replaces the fraction $\pi/(900 + 10\pi)$ of that input, approximately 0.34%. The table's 10% refers specifically to damping; the fraction of total hammer work replaced is much smaller here.

It can substitute for the input saving from reducing losses. On this sine motion, write $r = J\nu^2/k$. For $0 < r < 1$, X_r covers the fraction r of the damping work, giving the same hammer-input reduction as lowering b by that fraction. Actual heating remains W_b in the fictional model. At $r = 1$, it pays all the damping work; the inertial and spring torques also cancel on this particular motion, so the required hammer torque is zero throughout the segment. The initial rotation still has to be prepared. For $r > 1$, its supply exceeds the damping work, beyond what reducing a nonnegative damping coefficient alone can reproduce. Increasing b raises both supply and heating equally; changing J or k also changes the output mechanics. A larger X_r alone is not an optimum.

It can replace braking effort only during absorption. Where $\theta''' \Omega > 0$, X_r takes the instantaneous power $X_r \theta''' \Omega$. It can cover that amount of a separately required braking load; conventional braking must cover any shortfall. Excess absorption requires an adjusted drive or motion. Slowing down alone does not establish this role: the worked sine slows to rest while X_r supplies power, so it provides no replacement for a brake there.

Contact design and control still determine when the assistance is useful. The term has no independent timing command: the motion fixes its sign. Where $\theta''' = 0$, it provides no torque assistance; where $\Omega = 0$, it transfers no power. Its work contribution therefore does not specify a replacement for contact geometry, engagement and release, sensing, or the decision to stop at the tightening target. Nor does the local work saving specify a percentage reduction in hammer size or motor rating.

The quantified replacement concerns work at the output during a specified motion. Battery savings also depend on motor and hammer preparation and on whatever supplies or replenishes X_r . Its existence and coefficient alone leave those costs open.

References

1. Andreas Wettstein, Patric Grauberger and Sven Matthiesen (2021). [Modeling dynamic mechanical system behavior using sequence modeling of embodiment function relations: case study on a hammer mechanism](#). *SN Applied Sciences* **3**, article 128. DOI: 10.1007/s42452-021-04149-8.
2. Mark Benazet, Francesco Ricca, Dario Bralla, Melanie N. Zeilinger and Andrea Carron (2026). [Learning-based approximate model predictive control for an impact wrench tool](#). *European Journal of Control*, article 101596, available online 28 July 2026. DOI: 10.1016/j.ejcon.2026.101596.